

Lab 2: DT Signal Operations

Instructions:

1. Plots should include *x* label, *y* label, legend, and titles
2. If the code is found to be copied and you are not able to explain, zero marks will be awarded. **Ensure that you show the outputs of the task to the TAs and get them evaluated. No queries will be entertained post lab session.**

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1. In Lab 1, the **Toeplitz matrix** was introduced to you. The problem is repeated here, so you can recall and understand its use to obtain linear convolution.

When the sequences $x(n)$ and $h(n)$ have finite durations N_x and N_h , respectively, then their linear convolution can also be implemented using matrix-vector multiplication. If elements of $y(n)$ and $x(n)$ are arranged in column vectors x and y , respectively, then linear convolution can be obtained as $y = Hx$, where linear shifts in $h(n - k)$ for $n = 0, \dots, N_h - 1$ are arranged as rows in the matrix H . This matrix has an interesting structure and is called a **Toeplitz matrix**.

To investigate this matrix, consider the following sequences

$$x(n) = \{1, 2, 3, 4, 5\} \text{ and } h(n) = \{p, q, r, s\}$$

↑

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where p, q, r , and s are the last 4 digits of your ID, if any digit is zero, add 8 to that digit.

- a) What can you say about the first column and the first row of H ? [part d in Lab 1]

MATLAB provides a function called **toeplitz** to generate a Toeplitz matrix, given the first row and the first column.

Using this function and your answer to part a [part d in Lab 1], develop another MATLAB function to implement **linear convolution**. The format of the function should be as below, verify your function for the above sequence.

```
function [y,H]=conv_tp(h,x)
% Linear Convolution using Toeplitz Matrix
% -----
% [y,H]=conv_tp(h,x)
% y = output sequence in column vector form
% H = Toeplitz matrix corresponding to sequence h so that y = Hx
% h = Impulse response sequence in column vector form
% x = input sequence in column vector form
```

2. For question 1, now write a function which replicates the method we discussed in the lecture, where we used normal arithmetic multiplication to compute linear convolution. Your final answer should also indicate the position of terms after linear convolution. Do you see any similarity between this method and using Toeplitz to evaluate the linear convolution?
3. Obtain the area under the function $f(t) = \exp(-t)$ from 0 to 2 and for $T = 0.5$ and $T = 0.1$ using the help of MATLAB. [Hint: Use the idea of Digital Differentiator and Integrator discussed in Tutorial 2]

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4. Let $x(n) = [3, 11, a, 0, -1, b, 2]$ be a prototype sequence, $y(n)$ be its noise-corrupted-and-shifted version $y(n) = x(n - 2) + w(n)$, where $w(n)$ is Gaussian sequence with mean 0 and variance 1, the constants a and b are the last and second last digits of your id, if any digit is zero, add 6.
- a) Compute the cross-correlation between $y(n)$ and $x(n)$. Plot the result. [Without using inbuilt command]
 - b) Also, plot the autocorrelation of the prototype sequence. [without using inbuilt command]
 - c) Correlation can be implemented as convolution, the inbuilt command in MATLAB is **xcorr**, implement using both the methods and compare.
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